

ACTL3182 Wk2

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Set up: We have n risky assets with random returns $R_1, R_2, \dots, R_n$. Suppose we know the expected return vector and covariance matrix

$$\mathbf{z} = \begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_n \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \sigma_1^2 & \rho_{1,2}\sigma_1\sigma_2 & \dots & \rho_{1,n}\sigma_1\sigma_n \\ \rho_{2,1}\sigma_2\sigma_1 & \sigma_2^2 & \dots & \rho_{2,n}\sigma_2\sigma_n \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{n,1}\sigma_n\sigma_1 & \rho_{n,2}\sigma_n\sigma_2 & \dots & \sigma_n^2 \end{pmatrix}.$$

Problem: How can we construct a good portfolio?

Portfolio theory recap

We can represent a portfolio with a vector $\mathbf{w}$, where $\mathbf{w}_i$ represents the portion of the i th asset in our portfolio.

Let R_p be (random) return of our portfolio and $\mathbf{R}$ be the vector of (random) returns of all n assets. Then,

$$R_p = \mathbf{w}^\top \mathbf{R},$$

$$\mathbb{E}[R_p] = \mathbf{w}^\top \mathbf{z},$$

$$\text{Var}(R_p) = \mathbf{w}^\top \Sigma \mathbf{w}.$$

If $\mathbf{w}_A$ and $\mathbf{w}_B$ are weights of portfolios with random returns R_A and R_B , then

$$\text{Cov}(R_A, R_B) = \mathbf{w}_A^\top \Sigma \mathbf{w}_B.$$

Let $\mathbf{w}, \mathbf{v} \in \mathbb{R}^n$ and $\Sigma \in \mathbb{R}^{n \times n}$ be symmetric. Then,

$$\frac{\partial}{\partial \mathbf{w}} (\mathbf{w}^\top \Sigma \mathbf{w}) = 2\Sigma \mathbf{w}.$$

$$\frac{\partial}{\partial \mathbf{w}} (\mathbf{w}^\top \mathbf{v}) = \mathbf{v}.$$

Math recap (cont.)

Consider the optimisation problem:

$$\min_{\mathbf{w} \in \mathbb{R}^n} f(\mathbf{w}) \quad \text{subject to} \quad g(\mathbf{w}) = 0.$$

Then, the solution is the stationary point of the Lagrangian,

$$\mathcal{L}(\mathbf{w}, \lambda) = f(\mathbf{w}) + \lambda g(\mathbf{w}).$$

This can be extended to multiple constraints via

$$\mathcal{L}(\mathbf{w}, \lambda, \gamma) = f(\mathbf{w}) + \lambda g_1(\mathbf{w}) + \gamma g_2(\mathbf{w}).$$

Markowitz Mean-Variance Optimisation Problem

Markowitz

For a given fixed return $\mu > 0$, we seek the portfolio that achieves this return with minimal variance. That is, we seek the solution to

$$\min_{\mathbf{w} \in \mathbb{R}^n} \frac{1}{2} \mathbf{w}^T \Sigma \mathbf{w} \quad \text{subject to} \quad \mathbf{w}^T \mathbf{1} = 1, \quad \mathbf{w}^T \mathbf{z} = \mu,$$

or equivalently, the stationary point of

$$\mathcal{L}(\mathbf{w}, \lambda, \gamma) = \frac{1}{2} \mathbf{w}^T \Sigma \mathbf{w} + \lambda(1 - \mathbf{w}^T \mathbf{1}) + \gamma(\mu - \mathbf{w}^T \mathbf{z}).$$

Efficient Frontier

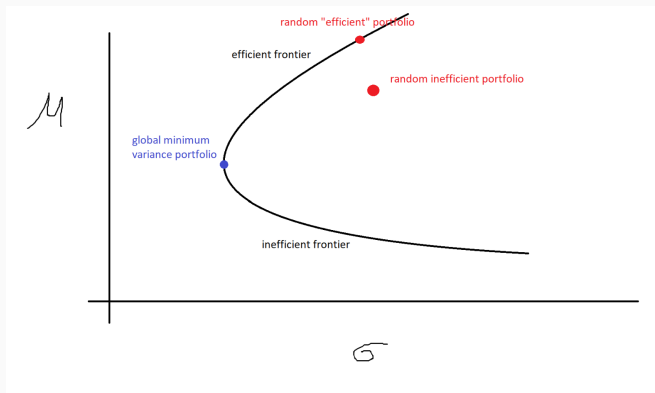


Figure 1: Efficient Frontier (credit Arkie)

Some results

Global Minimum Variance Portfolio

The global minimum variance portfolio is the solution to

$$\min_{\mathbf{w} \in \mathbb{R}^n} \frac{1}{2} \mathbf{w}^T \Sigma \mathbf{w} \quad \text{subject to} \quad \mathbf{w}^T \mathbf{1} = 1.$$

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Two Fund Theorem

Any minimum variance portfolio can be written via two distinct minimum variance portfolios, that is, if $\mathbf{w}$ is a MVP, for any $\mathbf{w}_A, \mathbf{w}_B$ we have $x \in \mathbb{R}$ such that

$$\mathbf{w} = x \mathbf{w}_A + (1 - x) \mathbf{w}_B.$$

Inclusion of Risk free asset

Suppose we have a risk free asset - the investor can invest or borrow money at a fixed rate r_f .

What changes?

- We no longer need $\mathbf{w}^T \mathbf{1} = 1$, for we can borrow/invest whatever more is needed.
- We want to be compensated for risk: if we are taking on more variance, we want to be compensated with an excess return over r_f .

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By the last point, we may seek to maximise the Sharpe ratio

$$S(\mathbf{w}) = \frac{\mu_p - r_f}{\sigma}.$$

Markowitz with risk-free asset

Markowitz with risk-free asset

For a given fixed return $\mu > 0$ we seek the portfolio that maximises the Sharpe ratio,

$$\max_{\mathbf{w} \in \mathbb{R}^n} \frac{(\mathbf{w}^T \mathbf{z} + (1 - \mathbf{w}^T \mathbf{1})r_f) - r_f}{\sigma_p}$$

subject to

$$\mathbf{w}^T \mathbf{z} + (1 - \mathbf{w}^T \mathbf{1})r_f = \mu,$$

or equivalently the stationary point of

$$\mathcal{L}(\mathbf{w}, \gamma) = \frac{1}{2} \mathbf{w}^T \Sigma \mathbf{w} + \gamma(\mu - r_f - (\mathbf{z} - r_f \mathbf{1})^T \mathbf{w}).$$

Tangency Portfolio and one-Fund theorem

Tangency Portfolio

The tangency portfolio is the above when $\mathbf{w}^T \mathbf{z} = 1$, that is, when the portfolio consists entirely of risky assets.

